Causal Representation Learning and Causal Discovery in Deep Learning: A Survey

Causal representation learning and causal discovery aim to make deep learning systems reason about mechanisms, interventions, counterfactuals, and distribution shifts rather than only correlations. Causal representation learning asks how latent variables can recover meaningful causal factors from high-dimensional observations, while causal discovery asks how to infer causal graphs or mechanisms from observational, interventional, or temporal data. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around causal representations, deep structure learning, invariance, counterfactual and interventional learning, temporal discovery, generative models, applications, identifiability, and evaluation. The survey argues that progress requires unifying statistical assumptions, representation learning, optimization, and domain knowledge.

Keywords:

causal representation learning; causal discovery; deep learning; causal graphs; counterfactuals; invariance; time series; identifiability

1 Introduction

Deep learning excels at prediction but often struggles under interventions, distribution shifts, and counterfactual queries. Causal learning addresses these limits by modeling how variables influence one another and how mechanisms remain stable across environments. Causal representation learning and causal discovery bring this agenda into high-dimensional settings where variables may be images, text, trajectories, embeddings, or latent states [78, 142, 155, 174, 238, 243, 284, 294].

Causal representation learning seeks latent factors that correspond to causal variables or mechanisms. Causal discovery seeks graph structure, temporal relations, or functional mechanisms. Deep learning enters both problems: it parameterizes nonlinear mechanisms, learns representations from raw data, optimizes differentiable graph objectives, and scales causal methods to complex domains. Yet deep models can also hide assumptions and produce spurious causal interpretations [47, 142, 155, 174, 238, 243, 284, 294].

This survey reviews the intersection of causal representation learning, causal discovery, and deep learning. We focus on assumptions, method families, evaluation, and open challenges.

2 Background

2.1 Causal Models and Deep Learning

Causal models describe how variables are generated by mechanisms and how interventions modify those mechanisms. Directed acyclic graphs, structural causal models, potential outcomes, and temporal causal models provide formal languages for reasoning about interventions and counterfactuals. Deep learning contributes flexible function approximators, but causal interpretation still depends on assumptions about interventions, independence, environments, and identifiability [47, 174, 241, 269, 274, 284, 294, 297].

2.2 Causal Representation Learning

Causal representation learning aims to recover high-level causal variables from observations. In images, causal factors may include objects, positions, lighting, or actions. In sequential domains, they may include states, goals, or mechanisms. The central difficulty is identifiability: many latent representations can explain the same observations unless additional assumptions, interventions, temporal structure, or inductive biases are used [47, 142, 155, 174, 238, 243, 284, 294].

2.3 Causal Discovery

Causal discovery infers graphs or mechanisms from data. Constraint-based methods test conditional independences; score-based methods optimize graph scores; differentiable methods relax acyclicity constraints for neural optimization; functional causal models exploit asymmetries in cause-effect mechanisms. Deep causal discovery extends these ideas to nonlinear, high-dimensional, temporal, or representation-learning settings [47, 174, 241, 269, 274, 284, 294, 297].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with causal representation learning, causal representations, disentanglement, latent causal models, independent causal mechanisms, causal world models, deep and neural causal discovery, differentiable causal discovery, causal structure learning, DAG learning, NOTEARS, DAG-GNN, GraN-DAG, causal inference, counterfactual representation learning, interventional data, treatment effects, invariant causal prediction, invariant risk minimization, domain generalization, out-of-distribution robustness, time-series causal discovery, Granger causality, reinforcement learning, vision, language, healthcare, biology, benchmarks, identifiability, structural causal models, causal generative models, and causal VAEs. Records were retained when title or abstract evidence connected causal representation, causal discovery, causal inference, invariance, interventions, counterfactuals, temporal discovery, or identifiability with machine learning or deep learning. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *observed data*: tabular, image, text, graph, temporal, reinforcement-learning trajectory, or multimodal. The second is *causal target*: latent variable, graph edge, mechanism, intervention effect, counterfactual, or invariant predictor. The third is *supervision*: observational, interventional, weakly supervised, multi-environment, temporal, or synthetic. The fourth is *model class*: structural causal model, variational latent model, graph neural network, transformer, normalizing flow, diffusion model, or reinforcement-learning world model. The fifth is *assumption*: acyclicity, faithfulness, sparsity, independent noise, modular mechanisms, temporal ordering, or environment invariance. The sixth is *optimization*: constraint-based testing, score search, differentiable relaxation, adversarial training, contrastive learning, or variational inference. The seventh is *evaluation*: graph recovery, intervention prediction, counterfactual accuracy, OOD generalization, identifiability, and downstream utility. The eighth is *application*: vision, language, healthcare, biology, robotics, recommendation, and scientific discovery.

3.3 Learning Causal Representations

Representation learning methods use inductive biases, temporal ordering, actions, interventions, sparsity, object structure, or multiple environments to make causal variables identifiable. Disentanglement alone is insufficient unless the learned factors align with mechanisms. Recent work therefore combines representation objectives with causal constraints, weak labels, interventions, or environment variation [47, 142, 155, 174, 238, 243, 284, 294].

3.4 Deep Structure Learning

Deep causal discovery methods parameterize nonlinear mechanisms and optimize graph structure. Differentiable acyclicity constraints, neural functional models, graph neural networks, and variational approaches allow end-to-end learning. However, graph recovery remains sensitive to sample size, hidden confounding, cyclic dynamics, measurement noise, and model misspecification [47, 174, 241, 269, 274, 284, 294, 297].

3.5 Invariance and Generalization

Invariance-based methods exploit the idea that causal mechanisms remain stable across environments while spurious correlations shift. Invariant risk minimization, domain generalization, and stable prediction methods search for features that transfer under interventions or distribution shifts. The challenge is defining environments and avoiding degenerate solutions that satisfy invariance without capturing causality [47, 111, 116, 149, 155, 160, 174, 262].

3.6 Counterfactual and Interventional Learning

Counterfactual learning asks what would have happened under a different action or treatment. Interventional data can identify causal effects and help learn mechanisms, but interventions may be expensive, partial, or confounded. Deep models are useful for heterogeneous treatment effects, policy evaluation, and high-dimensional counterfactuals, yet they require careful uncertainty and overlap assumptions [116, 120, 142, 155, 174, 238, 243, 284].

3.7 Temporal Causal Discovery

Time-series methods use lag structure, Granger causality, state-space models, and temporal attention to infer directed dependencies. Temporal ordering helps identifiability, but latent confounding, contemporaneous effects, nonstationarity, and irregular sampling remain difficult. Deep models can capture nonlinear dynamics while preserving temporal causal constraints [34, 78, 111, 131, 140, 241, 279, 294].

3.8 Generative Models and Mechanisms

Variational autoencoders, normalizing flows, diffusion models, and other generative architectures can model latent causal mechanisms and intervention effects. The key question is whether the learned latent space corresponds to intervenable variables. Causal generative models often combine structural assumptions with reconstruction, independence, or intervention losses [116, 120, 142, 155, 174, 243, 274, 284].

3.9 Applications

Causal deep learning is used in computer vision, language, reinforcement learning, healthcare, biology, recommender systems, robotics, and scientific discovery. Applications differ in available interventions, domain knowledge, temporal structure, and tolerance for causal error. In high-stakes domains, interpretability and uncertainty are as important as predictive accuracy [47, 77, 78, 118, 174, 241, 243, 299].

3.10 Evaluation and Identifiability

Evaluation is hard because ground-truth causality is rarely observed. Synthetic benchmarks provide known graphs but may be too simple. Real datasets provide realism but weak ground truth. Identifiability analysis clarifies when causal claims are possible, while benchmark design should test intervention prediction, counterfactual validity, graph recovery, and OOD transfer [47, 116, 142, 155, 243, 274, 284, 294].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across causal representations, deep causal discovery, invariance, counterfactuals, temporal discovery, generative models, applications, and evaluation.

Causal representation learning and disentanglement The causal representation learning and disentanglement cluster contains work that develops methods, systems, datasets, or analyses central to this survey [78, 118, 120, 174, 238, 240, 241, 243, 284]. A second strand in this cluster studies design tradeoffs, evaluation protocols, failure modes, and domain-specific variants [37, 117, 140, 178, 187, 214, 236, 253, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 26, 60, 73, 96, 112, 127, 218, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 18, 23, 137, 177, 248, 266, 273, 286]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [30, 36, 76, 101, 123, 128, 216, 235, 267]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [27, 75, 102, 150, 184, 194, 225, 265, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 62, 133, 191, 251, 256, 264, 282, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 237, 242].

Deep causal discovery and structure learning The deep causal discovery and structure learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [64, 110, 163, 167, 175, 232, 274, 279, 297]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [25, 45, 63, 113, 138, 154, 230, 246, 280]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 21, 44, 107, 176, 192, 220, 239, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [16, 72, 83, 109, 183, 196, 199, 204, 231]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [40, 67, 105, 139, 159, 170, 188, 207, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 7, 42, 91, 156, 158, 208, 228, 263]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [32, 49, 54, 90, 169, 233, 255, 259, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 206, 209, 222].

Invariance, domain generalization, and robustness The invariance, domain generalization, and robustness cluster contains work that develops methods, systems, datasets, or analyses central to this survey [8, 52, 53, 59, 106, 135, 148, 229, 271]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [4, 19, 31, 97, 108, 157, 198, 203, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [93, 98, 99, 162].

Counterfactual and interventional learning The counterfactual and interventional learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [68, 88, 94, 125, 160, 173, 189, 197, 261].

Temporal and time-series causal discovery The temporal and time-series causal discovery cluster contains work that develops methods, systems, datasets, or analyses central to this survey [29, 66, 81, 86, 111, 119, 212, 250, 294]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [22, 48, 95, 124, 132, 182, 249, 276, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 17, 28, 65, 71, 134, 168, 181, 245]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [20, 82, 121, 144, 179, 202, 247].

Generative models and causal mechanisms The generative models and causal mechanisms cluster contains work that develops methods, systems, datasets, or analyses central to this survey [35, 70, 84, 104, 142, 155, 164, 223, 272]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [50, 51, 80, 114, 129, 136, 205, 210, 215]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [146, 166, 190, 260].

Applications in vision, language, reinforcement learning, and science The applications in vision, language, reinforcement learning, and science cluster contains work that develops methods, systems, datasets, or analyses central to this survey [2, 185, 295].

Benchmarks, identifiability, and evaluation The benchmarks, identifiability, and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [34, 47, 77, 116, 149, 200, 262, 269, 299]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [11, 69, 131, 143, 145, 151, 186, 201, 213]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [39, 43, 61, 103, 141, 161, 211, 217, 257]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 15, 55, 89, 180, 224, 227, 234, 287]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 87, 92, 126, 130, 252, 268, 277, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [79, 100, 147, 152, 165, 171, 195, 244, 254]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 41, 56, 74, 85, 153, 172, 221, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [33, 38, 115, 193, 219, 226, 270, 283, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 58, 122, 258, 300].

4 Challenges and Prospects

4.1 Identifiability in High-Dimensional Domains

Deep models can fit many latent explanations. Future work needs clearer assumptions, richer interventions, temporal and environmental structure, and diagnostics that expose when causal variables are not identifiable.

4.2 Hidden Confounding and Measurement

Real data often omit variables and measure proxies. Methods should model uncertainty about hidden causes and avoid presenting learned graphs as definitive when the observed variables are incomplete.

4.3 Interventions and Active Data Collection

Causal learning improves with interventions, but interventions are costly. Active learning, experimental design, and simulation can help choose informative interventions while respecting safety and resource constraints.

4.4 Causal Foundation Models

Large models may encode causal regularities from broad data, but they also encode correlations. Integrating causal reasoning with foundation models will require explicit variables, tools for intervention, and evaluation beyond text plausibility.

4.5 Evaluation Standards

The field needs benchmarks that test graph recovery, intervention prediction, counterfactual reasoning, OOD transfer, and robustness under hidden confounding. Reporting assumptions should be as important as reporting accuracy.

5 Conclusion

Causal representation learning and causal discovery bring causal reasoning into deep learning. They promise models that can generalize under shift, answer counterfactual questions, and support interventions, but they also depend on strong assumptions about mechanisms, environments, and identifiability. This survey reviewed the literature across causal representations, deep structure learning, invariance, interventional learning, temporal discovery, generative mechanisms, applications, and evaluation. The central challenge is to make causal claims both learnable and auditable in realistic high-dimensional systems.

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